Superpixel-Based Multiple Statistical Feature Extraction Method for Classification of Hyperspectral Images

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***Abstract*— To improve the classification accuracies of hyperspectral images (HSIs), especially when using a limited number of training samples, a novel superpixel-based multiple statistical feature extraction (SPMSFE) method is proposed in this article. For each dimension-reduced pixel obtained by maximum noise fraction (MNF), the most similar superpixel-based neighbors of different sizes are first identified based on the spatial structures of the HSIs to exploit contextual spatial information accurately. Then, multiple statistical features, including the mean, covariance descriptor, and the Gaussian feature, are extracted for the set of superpixel-based neighbors to fully explore the spatial geometry information, tight correlations between different spectral bands, and spatial–spectral variations from different perspectives, respectively. In addition, these three statistical features of the pixels share the same size and can be utilized for uniform classification without any dimensionality obstacles even though the sizes of the superpixel-based neighbors for different pixels may be different. Next, we construct multiple kernels to map these multiple statistical features in the Euclidean and Riemannian manifold spaces to a uniform Hilbert space and embed them into a multitask kernelized sparse representation classification (MTKSRC) model. The constructed MTKSRC model provides a natural method to effectively fuse the multiple statistical features for excellent classification performance and robustness, especially when using limited numbers of training samples. The experimental results for three widely used HSI data sets demonstrate that the classification accuracy of the proposed SPMSFE method outperforms several latest and state-of-the-art classification methods by a large margin.**

***Index Terms*— Covariance descriptor, feature extraction (FE), Gaussian model, hyperspectral classification, mean vector, sparse representation (SR) classifier.**

I. Introduction

HYPERSPECTRAL images (HSIs) are composed of data from several hundreds of adjacent and narrow spectral bands to increase the possibility of discriminating land covers accurately; hence, they have been successfully used in many remote sensing applications, such as spectral unmixing [1]–[4], military target detection [5]–[7], land-cover analysis [8]–[10], and scene classification [11]–[14]. Among these research fields, classification aimed at assigning each test pixel to a unique class is important because of its implications in urban investigations, precision agriculture, and environmental monitoring. To achieve such classification, various classifiers have been developed for HSIs, including support vector machines (SVMs) [15]–[17], multinomial logistic regression (MLR) [18]–[20], sparse representation (SR) [21]–[23], and neural networks [24]–[27]. However, the existing problems of HSI classifications are mainly attributed to two aspects: the first is that the classification accuracy degrades significantly when there are very limited training samples (Hughes phenomenon [28]), and the second is that the classification maps are always degraded by noise due to the existence of mixed spectral pixels.

To address these two problems, many feature extraction (FE) methods [29]–[34] have been proposed. The objective of these FE methods is to incorporate various discriminative spatial and spectral information for HSI classification to enlarge the separability among the pixels. The FE methods can be divided into two categories as follows: handcrafted FE methods and deep-learning-based FE methods. In [35], the extended morphological profile (EMP) using morphological opening and closing operations is exploited to extract spatial information adaptively, which can improve the classification accuracy significantly. Inspired by the success of the EMP method, a series of related approaches were developed [36]–[39]. In [40], the authors employed local binary patterns (LBPs) to extract the contextual spatial features and improve classification performance. Moreover, a number of multiple spectral–spatial FE methods are presented in [41]–[44] to sufficiently describe the pixels from different perspectives, where multiple kernel learning [45]–[48] can be embedded to further enhance classification accuracy. Based on the assumption that the neighboring pixels are more likely to belong to the same class, the spatial context information within a predefined square window is extracted using the joint sparse representation classification (JSRC) model in [21]. As the JSRC does not fully consider the spatial distribution of HSI, several superpixel-based methods were developed in [49]–[52] to extract the spatial information adaptively from a relatively

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large window for homogeneous areas and a relatively small window for heterogeneous areas. Although these handcrafted FE methods provide significant improvement for HSI classification, they depend heavily on the number of training samples. Extracting more discriminative features to improve classification performance when constrained by a limited number of training samples is, thus, considered a challenging task.

Motivated by the successful application of deep learning [53]–[55] in many computer vision tasks, it has been widely utilized for HSI classification as it is capable of extracting not only spatial but also spectral features. Compared to the handcrafted FE methods, deep learning can automatically extract both simple and complex features via a series of hierarchical layers from the original HSI data sets. In [25], the authors introduced a convolutional neutral network (CNN) for the spectral domain for HSI classification. In [56], deep spatial features extracted using a CNN were encoded in low-dimensional sparse features to enhance feature representation and improve classification accuracy. Then, a number of CNN-based methods [13], [57], [58] were proposed to extract both spectral and spatial features to further improve classification performance. In [59], a deep feature fusion network was presented to mitigate the overfitting problem of CNN-based methods by exploiting residual learning, which achieves superior classification accuracy compared to conventional methods. Although deep-learning-based FE methods can be used to explore spatial–spectral information automatically for HSI classification, there are two main flaws. The first is that many supervised methods require a large number of samples to train the deep networks, while the available training samples in HSI classification are usually very limited, thereby rendering unsatisfactory classification performance, especially for limited numbers of training samples. The second is that a large number of parameters in the deep networks need to be tuned, which always results in high computational complexity.

Recently, several researchers have considered extracting effective statistical features, such as the simple mean vector and the second-order covariance matrix (CM), to improve classification performance when training with limited samples. In [49], the mean and weighed features within and among the superpixels were exploited to explore the spatial information to achieve good classification accuracy. Moreover, the CM, whose nondiagonal entries represent the correlations between two different features, is one of the most effective statistical features for classification. In [32], an FE method using the local CM representation was proposed to fully characterize the correlations among different spectral bands and spatial information, which outperformed several state-of-the-art approaches when training with limited samples. Then, an FE method based on multiscale covariance maps (MCMs) was developed for CNN-based HSI classification in [31]. Nevertheless, utilizing either the simple mean feature or second-order CM can only characterize the data set from one perspective, and it is difficult to obtain robust classification accuracy. Gaussian models have been commonly known to adequately estimate the mean vector and CM, thus providing a method to simultaneously represent the characteristics of the mean and covariance features. However, the most important challenge is that some regions of the HSI data sets may not always follow the Gaussian distribution and do not satisfy the assumptions of the Gaussian model.

In this article, we propose a novel superpixel-based multiple statistical feature extraction (SPMSFE) method for the classification of HSIs. First, the most similar superpixel-based neighbors with different sizes are identified based on the spatial structure of the HSI for each dimension-reduced pixel obtained by maximum noise fraction (MNF), which not only accurately exploits the spatial context information but also reduces the computational complexity remarkably. Then, we calculate the mean vector, CM, and Gaussian distribution among the set of neighboring pixels as their multiple statistical features to characterize each pixel. These three statistical features can fully characterize the spatial geometry information, tight correlations between different spectral bands, and spatial–spectral variations from different perspectives, respectively, while still being complimentary with each other. Another advantage of the multiple statistical features is that they share the same sizes for different pixels and can be utilized for uniform classification even though the sizes of the superpixel-based neighbors for each pixel may be different. Next, we construct multiple kernels to map the multiple statistical features to a uniform Hilbert space since the mean feature is present in the Euclidean space, while the covariance descriptor and Gaussian features are in the Riemannian manifold space. Finally, we embed the multiple kernels and construct a multiple-task kernelized sparse representation classification (SRC) model, which offers a natural method of fusing the multiple statistical features for significantly improving the classification accuracy with enhanced robustness, especially for limited training samples. Experiments were performed on three widely used HSI data sets, and the results demonstrate that the proposed method achieves a much higher classification accuracy and outperforms several state-of-the-art classification methods by a large margin.

The major contributions of this article are summarized as follows.

1) Superpixel-based neighbors were identified based on spatial structures of the HSIs for each dimension-reduced pixel, which is beneficial for exploring the spatial context information accurately.
2) Multiple statistical features were extracted, including the mean feature, covariance descriptor, and the Gaussian feature, to fully exploit the spatial geometry information, correlations between different spectral bands, and spatial–spectral variations from different perspectives, respectively, to improve the classification accuracy with more robustness, especially for limited numbers of training samples.
3) Multiple kernels were designed to map the multiple statistical features to a uniform Hilbert space and construct a multiple-task kernelized SRC model, thus offering a natural method to fuse the multiple statistical features and further improve classification performance.

The remainder of this article is organized as follows. Section II presents a review of the related works, including

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the Gaussian distribution embedding and the log–Euclidean distance (LED). The details of the proposed SPMSFE method are presented in Section III, including superpixel-based neighbor identification, multiple statistical FE, multiple statistical feature-based multitask kernelized sparse representation classification (MTKSRC) modeling, and optimization and classification. The experimental results from three benchmark HSI data sets are described and analyzed in Section IV. Section V presents the conclusions of this work.

II. Related Works

A. Gaussian Distribution Embedding

The Gaussian distribution is a commonly used probability distribution that can simultaneously capture the mean and variations of data sets. As shown in [60], the Gaussian distribution can be embedded in a Riemannian manifold space. Accordingly, three main steps are considered as follows.

The first step involves characterization of the Gaussian distribution $\mathcal{N}(\boldsymbol{m}, \mathbf{C})$ by affine transformation $(\boldsymbol{m}, \mathbf{Q})$, where $\boldsymbol{m}$ is the mean vector, $\mathbf{C}$ is the CM, and $\mathbf{Q}$ is computed by decomposing $\mathbf{C}$ as $\mathbf{C} = \mathbf{Q}\mathbf{Q}^{\top}$.

The second step maps the affine group $\{(\boldsymbol{m}, \mathbf{Q})/\boldsymbol{m} \in \mathbb{R}^d, \mathbf{Q} \in \mathbb{R}^{d\times d}, |\mathbf{Q}| > 0\}$ into the simple Lie group $\{\mathbf{V}/\mathbf{V} \in \mathbb{R}^{(d+1)\times(d+1)}, |\mathbf{V}| > 0\}$ using the following equation:

$$(\boldsymbol{m}, \mathbf{Q}) \mapsto \mathbf{V} = |\mathbf{Q}|^{-1/(d+1)} \begin{bmatrix} \mathbf{Q} & \boldsymbol{m} \\ \boldsymbol{m}^{\top} & 1 \end{bmatrix} \tag{1}$$

where $|\mathbf{Q}|$ denotes the determinant of $\mathbf{Q}$.

The third step involves the further mapping of simple Lie group to a Riemannian manifold space according to the operation $\mathbf{V}\mathbf{V}^{T}$.

Finally, the d-dimensional Gaussian distribution $\mathcal{N}(\boldsymbol{m}, \mathbf{C})$ is embedded in the Riemannian manifold space and represented by a symmetric positive-definite (SPD) matrix $\mathbf{Y}$ with size $(d+1) \times (d+1)$, as shown in the following equation:

$$\mathcal{N}(\boldsymbol{m}, \mathbf{C}) \sim \mathbf{Y} = |\mathbf{Q}|^{-2/(d+1)} \begin{bmatrix} \mathbf{Q}\mathbf{Q}^{\top} + \boldsymbol{m}\boldsymbol{m}^{\top} & \boldsymbol{m} \\ \boldsymbol{m}^{\top} & 1 \end{bmatrix}. \tag{2}$$

B. Log–Euclidean Distance

For SPD matrices in the Riemannian manifold space, the common Euclidean-based distance is not suitable for measurement. The most widely utilized distance measures for the SPD matrices are the Stein divergence-based distance (SDD) [61], affine-invariant distance (AID) [62], and LED [63]. Since the LED not only induces a positive-definite kernel but also yields true geodesic distances, it is used in this work.

The LED is derived using the operation $\mathbf{C}_i \odot \mathbf{C}_j = \exp(\log(\mathbf{C}_i) + \log(\mathbf{C}_j))$, where $\mathbf{C}_i$ and $\mathbf{C}_j$ are two SPD matrices, and $\exp(\cdot)$ and $\log(\cdot)$, respectively, denote the common exponential and logarithm operators. The LED between $\mathbf{C}_i$ and $\mathbf{C}_j$ is then defined in the domain of matrix logarithms, as shown in the following equation:

$$d(\mathbf{C}_i, \mathbf{C}_j) = \|\log(\mathbf{C}_i) - \log(\mathbf{C}_j)\|_F \tag{3}$$

where $\|\cdot\|_F$ represents the Frobenius norm.

TABLE I
Notations and Definitions

Notations	Definitions
d	Original spectral dimensionality
l	Dimensionality reduced by MNF
L	Number of superpixels
M	Threshold of superpixel based neighbours
K	Sparsity level
ϵ	Minimum error limit

III. Proposed Method

Fig. 1 shows the schematic of the proposed SPMSFE method for the HSI classification. In the following section, we describe the four main steps of the proposed SPMSFE method in detail. Table I lists the important notations and definitions.

A. Identification of Superpixel-Based Neighbors

We identify the most similar superpixel-based neighbors for each pixel to explore the spatial–spectral information accurately. The detailed steps are as follows.

1) *Dimensional Reduction:* Let $\mathcal{X} \in \mathbb{R}^{s_1\times s_2\times d}$ be the original HSI data set; we then implement MNF on it to obtain the dimension-reduced data $\mathcal{F} \in \mathbb{R}^{s_1\times s_2\times l}$. Dimensional reduction using the MNF not only reduces the computational complexity greatly but also effectively removes the noise in the original HSI.
2) *Generation of 2-D Superpixel Map:* Compared to the local square windows, the sizes and shapes of the superpixels can be changed adaptively based on different spatial structures. The steps to generate the 2-D superpixel map are summarized. First, the principal component analysis (PCA) is applied to the original HSI data set to obtain the base images (i.e., the first three principal components) and reduce computational complexity. Second, we choose L according to the complexity of the structural textures in the HSI, which is reflected by implementing a texture detector (e.g., Sobel filter [64]) on the base image. Then, the oversegmentation algorithm [65] is applied to the base image to generate the 2-D superpixel map. Fig. 2 shows the 2-D superpixel map of the Indian Pines data set when L is fixed as 100, with the corresponding enlarged part shown in the black rectangle. As can be observed from Fig. 2, the sizes of the superpixels are relatively large for the homogeneous areas and relatively small for the heterogeneous areas. However, some superpixels may stride the edge and be clustered with pixels belonging to other classes, as shown in Fig. 2. To address this problem, further refining of similar neighbors is necessary.
3) *Identification of Superpixel-Based Neighbors:* First, the 3-D superpixel for each pixel is obtained by applying the indexes of the 2-D superpixel map to the dimension-reduced data $\mathcal{F} \in \mathbb{R}^{s_1\times s_2\times l}$. Each 3-D superpixel is then represented by $\mathbf{F}_i = [\boldsymbol{f}_i^1, \boldsymbol{f}_i^2, \ldots, \boldsymbol{f}_i^n], i = 1, 2, \ldots, L$, where n denotes the number of pixels. Then, to avoid introducing dissimilar neighbors, we refine the

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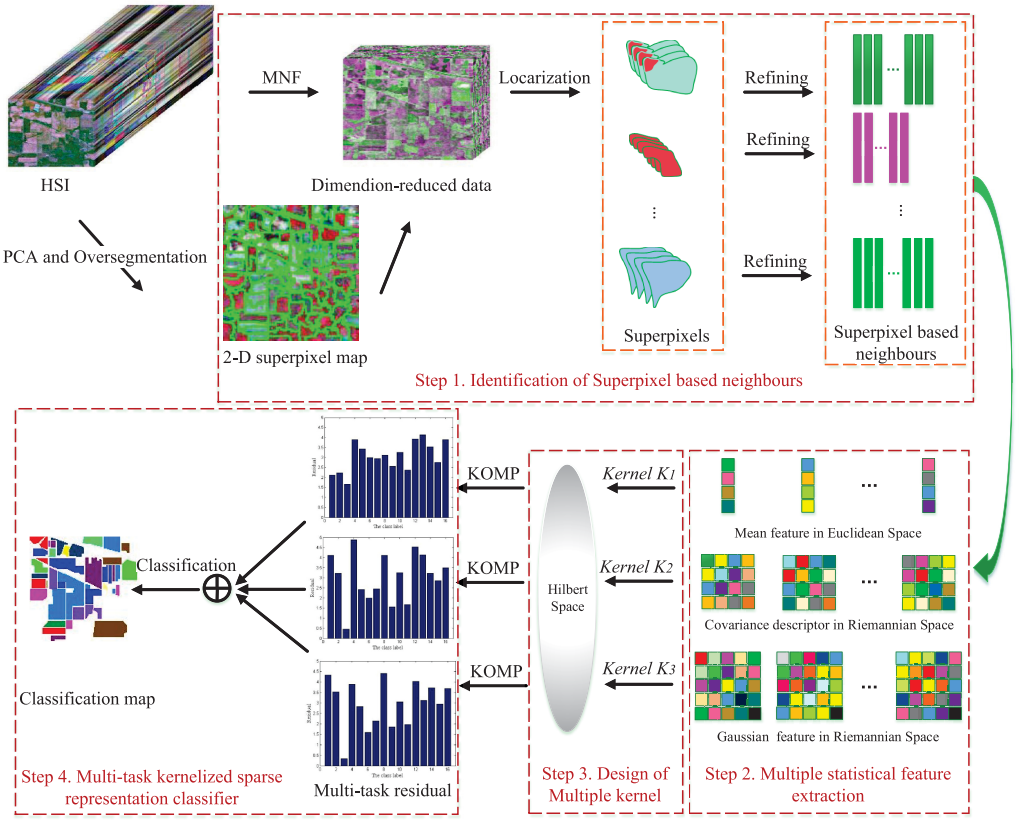


Fig. 1. Schematic of our proposed method SPMSFE for HSI classification.

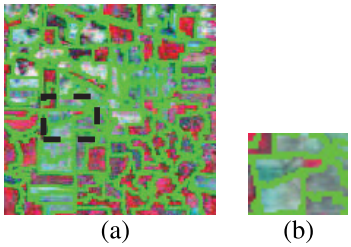


Fig. 2. Example of 2-D superpixel map of Indian Pines data set. (a) 2-D superpixel map. (b) Enlarged part corresponding to the black rectangle.

most similar neighbors from the corresponding superpixel for each pixel. The principle of identification is that if the number of pixels within a superpixel is greater than M, we select the $M-1$ most similar neighboring pixels based on cosine distance, which is calculated using the following equation:

$$\cos\left(\boldsymbol{f}_i^1, \boldsymbol{f}_i^j\right) = \frac{\left\langle \boldsymbol{f}_i^1, \boldsymbol{f}_i^j \right\rangle}{\left\| \boldsymbol{f}_i^1 \right\|_2 \left\| \boldsymbol{f}_i^j \right\|_2}, \quad j = 1, 2, \ldots, n. \tag{4}$$

However, if the number of pixels within a superpixel is smaller than M, all the pixels are selected as the superpixel-based neighbors. Consequently, the sizes of the superpixel-based neighbors may differ for the different pixels based on their spatial structures, which is beneficial to fully exploit the spatial context information accurately. In the following sections, we use M' to represent this variable value (equal to or smaller than M).

B. Multiple Statistical Feature Extraction

To deeply mine the spatial–spectral information and correlations from different perspectives for HSI classification, we extract the multiple statistical features, including the mean feature, covariance descriptor, and the Gaussian feature, which are sample-, subspace-, and distribution-based features, respectively, that are complementary to each other and can adequately describe the pixels. The details of the FE are as follows.

Mean Feature: The set of superpixel-based neighbors for each pixel is represented as $\mathbf{F}_i = [\boldsymbol{f}_i^1, \boldsymbol{f}_i^2, \ldots, \boldsymbol{f}_i^{M'}] \in \mathbb{R}^{l \times M'}$. For each pixel, the mean among its superpixel-based neighbors $\mathbf{F}_i = [\boldsymbol{f}_i^1, \boldsymbol{f}_i^2, \ldots, \boldsymbol{f}_i^{M'}]$ is computed using the following equation:

$$\boldsymbol{m}_i = \frac{1}{M'} \sum_{j=1}^{M'} \boldsymbol{f}_i^j. \tag{5}$$

The mean feature measures the average position and tendency, which can be used to explore the simple spatial information within the superpixels.

Covariance Descriptor: For each pixel, the covariance descriptor is calculated among its set of superpixel-based neighbors $\mathbf{F}_i = [\boldsymbol{f}_i^1, \boldsymbol{f}_i^2, \ldots, \boldsymbol{f}_i^{M'}]$ using the following

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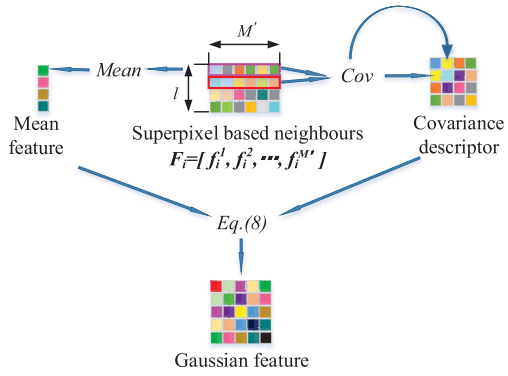


Fig. 3. Graphical illustration of the multiple statistical feature extraction for each pixel.

equation:

$$\mathbf{C}_i = \frac{1}{M'-1}\sum_{j=1}^{M'}\left(\boldsymbol{f}_i^j - \boldsymbol{m}_i\right)\left(\boldsymbol{f}_i^j - \boldsymbol{m}_i\right)^\top. \tag{6}$$

Each nondiagonal entry in the covariance descriptor represents the correlation between two different spectral bands to fully extract the correlations between different spectral bands containing important discriminative information for HSI classification.

Gaussian Feature: Since the HSI data set usually covers large variations, we exploit a Gaussian distribution model to deeply mine the spatial–spectral variances and correlations using the following equation:

$$g(\mathbf{F}_i) = \mathcal{N}\left(\mathbf{F}_i \middle/ \boldsymbol{m}_i^*, \mathbf{C}_i^*\right) \tag{7}$$

where $g(\mathbf{F}_i)$ is a Gaussian component with mean $\boldsymbol{m}_i^*$ and CM $\mathbf{C}_i^*$, which can be embedded in a Riemannian manifold space and represented by the SPD matrix $\mathbf{Y}_i$, as shown in the following equation:

$$\mathcal{N}\left(\boldsymbol{m}_i^*, \mathbf{C}_i^*\right) \sim \mathbf{Y}_i = |\mathbf{Q}_i|^{-2/(l+1)}\begin{bmatrix}\mathbf{Q}_i\mathbf{Q}_i^\top + \boldsymbol{m}_i^*\boldsymbol{m}_i^{*\top} & \boldsymbol{m}_i^* \\ \boldsymbol{m}_i^{*\top} & 1\end{bmatrix} \tag{8}$$

where $\mathbf{Q}_i$ is computed by decomposing the CM $\mathbf{C}_i^*$ as $\mathbf{C}_i^* = \mathbf{Q}_i\mathbf{Q}_i^\top$. Finally, a series of SPD matrices $\mathbf{Y}_i$ is obtained to represent the Gaussian features for the HSI. The Gaussian features simultaneously represent the positions and variations of the HSI data sets, thus offering a natural method to deeply explore the spatial geometry information and spectral–spatial correlations from different perspectives.

The graphical illustration of the proposed multiple statistical FE for a single pixel is shown in Fig. 3. The mean feature is useful for smooth areas with small variations, while the covariance descriptor provides discriminative spatial–spectral information for areas with large variations and sizes; the Gaussian feature simultaneously estimates the mean and covariance to determine spatial–spectral information and correlations for areas with small sizes and large variations. Although the Gaussian distribution cannot always be applied to all regions of HSI data sets, the mean feature and covariance descriptor are better choices to characterize the data set. Hence, one of the most important contributions of this work is combining multiple statistical features to characterize the original HSI for more robust classification performance. In addition, another advantage is that the extracted multiple statistical features for different pixels share the same sizes and can be utilized for uniform classification without any dimensionality problem even though the sizes of the superpixel-based neighbors for each pixel may differ.

C. Modeling of Multiple Statistical Feature-Based Classification

As the multiple statistical features are effective for mining the spatial–spectral information and are complementary to each other, they can be utilized as spatial–spectral features for SRC. Let $\mathbf{Z}^r = [z_1^r, z_2^r, \ldots, z_n^r], r = 1, 2, 3$, represent the three statistical features; for each pixel $z^r, r = 1, 2, 3$, the SRC model in the corresponding feature space can be defined in the following equation:

$$z^r = \mathbf{D}^r\boldsymbol{\alpha}^r + \boldsymbol{\epsilon}^r \tag{9}$$

where $\mathbf{D}^r = [\mathbf{D}_1^r, \mathbf{D}_2^r, \ldots, \mathbf{D}_C^r] \in \mathbb{R}^{L\times N}$ represents the directory comprising all the training samples randomly selected in the rth feature space; N is the total number of training samples, C is the number of classes, and $\mathbf{D}_c^r \in \mathbb{R}^{L\times N_c}$ denotes the subdirectory with N_c training samples selected from the cth class in the rth feature space. $\boldsymbol{\alpha}^r$ represents the sparse representation of z^r in the rth feature space, and $\boldsymbol{\epsilon}^r$ denotes the noisy error.

However, the mean feature is present in the Euclidean space, while the covariance descriptor and Gaussian feature are both in the Riemannian manifold space. In addition, the pixels in the Euclidean space always tend to be linearly inseparable, while those in the Riemannian manifold space cannot serve as direct inputs for classical learning algorithms. To fuse these multiple statistical features, they must be mapped to a uniform Hilbert space in which the separability among the samples can be enlarged maximally. Let $\Phi^r(\cdot), r = 1, 2, 3$ denote the mapping operator for the multiple statistical features; then, the model of the kernelized SRC (KSRC) in the high-dimensional Hilbert space can be improved as the following equation:

$$\Phi^r(z^r) = \Phi^r(\mathbf{D}^r)\boldsymbol{\alpha}^r + \boldsymbol{\epsilon}^r. \tag{10}$$

Since the multiple statistical features are mapped to a uniform Hilbert space and are complementary to each other, we can construct an MTKSRC model to fuse them and compute the sparse representation $\boldsymbol{\alpha}^*$ based on the following equation:

$$\begin{aligned}\boldsymbol{\alpha}^{r*} = &\arg\min_{\boldsymbol{\alpha}^r} \|\Phi^r(z^r) - \Phi^r(\mathbf{D}^r)\boldsymbol{\alpha}^r\|_2^2 \\ &\text{s.t. } \|\boldsymbol{\alpha}^r\|_0 \le K\end{aligned} \tag{11}$$

where $\|\boldsymbol{\alpha}^r\|_0$ denotes the l_0 norm, and K is the predefined sparsity level. Moreover, the main part of the optimization model in (11) can be decomposed, as shown in the following

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equation:

$$\begin{aligned}&\|\Phi^r(z^r)-\Phi^r(\mathbf{D}^r)\boldsymbol{\alpha}^r\|_2^2\\&\quad=(\Phi^r(z^r)-\Phi^r(\mathbf{D}^r)\boldsymbol{\alpha}^r)^\top(\Phi^r(z^r)-\Phi^r(\mathbf{D}^r)\boldsymbol{\alpha}^r)\\&\quad=\Phi^r(z^r)^\top\Phi^r(z^r)-\Phi^r(z^r)^\top\Phi^r(\mathbf{D}^r)\boldsymbol{\alpha}^r\\&\qquad-\boldsymbol{\alpha}^{r\top}\Phi^r(\mathbf{D}^r)^\top\Phi^r(z^r)+\boldsymbol{\alpha}^{r\top}\Phi^r(\mathbf{D}^r)^\top\Phi^r(\mathbf{D}^r)\boldsymbol{\alpha}^r.\end{aligned}\tag{12}$$

Nevertheless, the concrete form of the mapping $\Phi^r(\cdot), r = 1, 2, 3$ always tends to be unknown as a prior. To address this problem, "kernel tricks" defining a kernel function is embedded in the feature space. Since the multiple statistical features are in different spaces, we construct multiple kernel functions to map them to the high-dimensional Hilbert space.

As the mean feature lies in the Euclidean space, there are several well-studied kernels that can effectively map this data to the high-dimensional Hilbert space, such as the linear, the Gaussian radial basis function, and polynomial kernels. In this study, we utilize the linear kernel for the Hilbert embedding to fully utilize the Euclidean distance measure with rich representation, which can be expressed as

$$k_1(\boldsymbol{m}_i,\boldsymbol{m}_j)=\boldsymbol{m}_i^\top\boldsymbol{m}_j\tag{13}$$

where $\boldsymbol{m}_i$ and $\boldsymbol{m}_j$ are two arbitrary mean features.

As the covariance descriptor is actually an SPD matrix, we chose the LED as the distance measure. To embed the covariance descriptor in the Hilbert space, the Riemannian kernel can be exploited by computing the corresponding inner product in the Riemannian manifold space, as shown in the following equation:

$$k_2(\mathbf{C}_i,\mathbf{C}_j)=\text{trace}(\log(\mathbf{C}_i),\log(\mathbf{C}_j)).\tag{14}$$

As the Gaussian distribution is embedded in the Riemannian manifold space and represented using an SPD matrix of size $(l+1)\times(l+1)$, we employ the same Riemannian kernel to map the Gaussian feature to the Hilbert space. The Riemannian kernel for the two Gaussian features $\mathbf{Y}_i$ and $\mathbf{Y}_j$ in the Riemannian manifold space can be exploited, as shown in the following equation:

$$k_3(\mathbf{Y}_i,\mathbf{Y}_j)=\text{trace}(\log(\mathbf{Y}_i),\log(\mathbf{Y}_j)).\tag{15}$$

Then, the multiple statistical feature-based MTKSRC model with the embedded multiple kernels can be represented as follows:

$$\begin{aligned}\boldsymbol{\alpha}^{r*}&=\arg\min_{\boldsymbol{\alpha}^r}\{k_r(z^r,z^r)-k_r(z^r,\mathbf{D}^r)\boldsymbol{\alpha}^r\\&\qquad-\boldsymbol{\alpha}^{r\top}k_r(\mathbf{D}^r,z^r)+\boldsymbol{\alpha}^{r\top}K_r(\mathbf{D}^r,\mathbf{D}^r)\boldsymbol{\alpha}^r\}\\&=\arg\min_{\boldsymbol{\alpha}^r}\{k_r(z^r,z^r)-2k_r(z^r,\mathbf{D}^r)\boldsymbol{\alpha}^r\\&\qquad+\boldsymbol{\alpha}^{r\top}K_r(\mathbf{D}^r,\mathbf{D}^r)\boldsymbol{\alpha}^r\}\quad\text{s.t. }\|\boldsymbol{\alpha}^r\|_0\le K.\end{aligned}\tag{16}$$

Finally, the multiple statistical feature-based MTKSRC model offers a natural mapping method to the uniform Hilbert space to effectively fuse the multiple statistical features in the different spaces; this can improve classification accuracy significantly, with more robustness, especially with limited training samples.

Algorithm 1 KOMP

Require:
Test pixel z^r in statistical feature space; directory with training samples $\mathbf{D}^r$; corresponding kernel function $k_r(\cdot,\cdot)$; sparsity level K; minimum error limit ϵ.

Ensure:
1: Calculate $k_r(z^r,z^r)\in\mathbb{R}^1$, $K_r(\mathbf{D}^r,\mathbf{D}^r)\in\mathbb{R}^{N\times N}$, and $\mathcal{K}=k_r(z^r,\mathbf{D}^r)\in\mathbb{R}^N$;
2: Initialize the sparse representation $\mathbf{A}=zeros(N,1)$, set of indexes $\boldsymbol{\Delta}_0=\emptyset$, and residual error $\|\boldsymbol{rr}_t\|^2=k_r(z^r,z^r)$;
3: **while** $t\le K$ or $\|\boldsymbol{rr}_t\|^2\ge\epsilon^2$ **do**
4: Calculate $k_r(z^r,\mathbf{D}^r)-\boldsymbol{\alpha}^\top K_r(:,\boldsymbol{\Delta}_t)$ and choose the index id_t with the maximum value;
5: $\boldsymbol{\Delta}_t=[\boldsymbol{\Delta}_{t-1}\ id_t]$;
6: Compute $K_t=K[\boldsymbol{\Delta}_t,\boldsymbol{\Delta}_t]$ and $\mathcal{K}_t=\mathcal{K}[\boldsymbol{\Delta}_t]$;
7: Update sparse representation by $\boldsymbol{\alpha}_t=K_t^{-1}\mathcal{K}_t$;
8: Update the residual error by $\|\boldsymbol{rr}_t\|^2=k_r(z^r,z^r)-\mathcal{K}_t^\top\boldsymbol{\alpha}_t$;
9: **end while**

Output: sparse representation $\boldsymbol{\alpha}^r$

D. Optimization and Classification

To solve the optimization problem in (16) and obtain the sparse representation $\boldsymbol{\alpha}^{r*}$, the kernelized version of the orthogonal matching pursuit (KOMP) is used, as shown in Algorithm 1.

After obtaining the sparse representation $\boldsymbol{\alpha}^{r*}$ in the Hilbert space, the reconstruction error of each class is computed using

$$\begin{aligned}\boldsymbol{r}_i(z)&=\sum_{r=1}^{3}\left\|\Phi^r\left(z_i^r\right)-\Phi^r\left(\mathbf{D}_i^r\right)\boldsymbol{\alpha}_i^r\right\|_2^2\\&=\sum_{r=1}^{3}k_r\left(z_i^r,z_i^r\right)-2k_r\left(z_i^r,\mathbf{D}_i^r\right)\boldsymbol{\alpha}_i^r\\&\qquad+\boldsymbol{\alpha}_i^{r\top}K_r\left(\mathbf{D}_i^r,\mathbf{D}_i^r\right)\boldsymbol{\alpha}_i^r.\end{aligned}\tag{17}$$

Finally, the label of z is determined using the following equation:

$$\text{label}(z)=\arg\min_{i=1,2,\ldots,C}\boldsymbol{r}_i(z).\tag{18}$$

The detailed steps of the proposed method SPMSFE for HSI classification are presented in Algorithm 2.

IV. Experimental Results and Analysis

A. Data Set Description

In our experiments, the three benchmark HSI data sets used are the Indian Pines, University of Pavia, and Salinas data sets.

1) Indian Pines: The Indian Pines data set was gathered using the airborne visible/infrared imaging spectrometer (AVIRIS) from Northwestern Indiana. The size of this data set is $145\times145\times220$, with a spatial resolution of 20 m and spectral range of 0.4 to 2.5 μm. For the classification experiments, 20 noisy spectral bands (104th–108th, 150th–163rd, and 220th) are removed. This data set consists of 16 ground-truth classes, as shown in Table II. The false-color image and ground-truth map of the Indian Pines data set are shown in Fig. 4.

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Algorithm 2 SPMSFE for HSI Classification

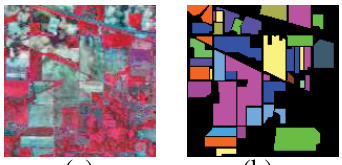
Require:

The original HSI data set $\mathcal{X}$; number of dimensions l; number of superpixels L; threshold of the superpixel-based neighbors M.

Ensure:

1: Employ MNF on $\mathcal{X} \in \mathbb{R}^{s_1 \times s_2 \times d}$ to obtain the dimension-reduced data $\mathcal{F} \in \mathbb{R}^{s_1 \times s_2 \times l}$;
2: Generate 2-D superpixel map;
3: Identify the most similar superpixel-based neighbors for each pixel as $\mathbf{F}_i = [\boldsymbol{f}_i^1, \boldsymbol{f}_i^2, \ldots, \boldsymbol{f}_i^{M'}] \in \mathbb{R}^{l \times M'}$;
4: Extract the mean feature $\boldsymbol{m}_i$, covariance descriptor $\mathbf{C}_i$, and Gaussian mixture feature $\mathbf{Y}_i$ for each pixel;
5: Utilize these three statistical features as the spatial–spectral features $\mathbf{Z}^r = [z_1^r, z_2^r, \ldots, z_n^r], r = 1, 2, 3.$;
6: Design the multiple kernels k_r using (13), (14), and (15) to map the multiple statistical features to the high-dimensional Hilbert space;
7: Randomly select N training samples to construct the directory $\mathbf{D}^r$ in the statistical feature space;
8: **for** each test pixel $z^r, r = 1, 2, 3.$ **do**
9: Obtain the sparse representation $\boldsymbol{\alpha}^r$ based on Algorithm 1;
10: Calculate the reconstruction error of each class in a multitask model using (17);
11: Determine the class label of z^r using (18).
12: **end for**

Output: The result classification map

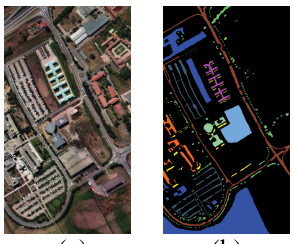


(a) (b)

Fig. 4. Indian Pines. (a) False-color image. (b) Ground-truth map.

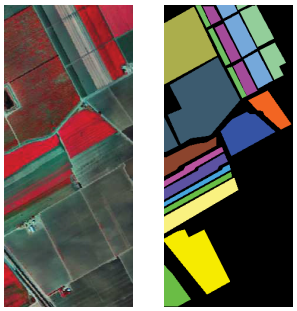
2) University of Pavia: The University of Pavia data set comprises images of the urban area surrounding this university in Italy and was acquired with the reflective optics system imaging spectrometer (ROSIS) sensor. The size of this data set is 610 × 340 × 115, with a spatial resolution of 1.3 m and spectral range of 0.43 to 0.86 μm. A total of 12 noisy spectral bands were removed from this set for the classification experiments. The data comprise nine ground-truth classes, which are shown in Table II. The false-color image and ground-truth map of the University of Pavia data set are shown in Fig. 5.

3) Salinas: The third data set consists of images of the Salinas Valley in California, USA, that was acquired with the ROSIS sensor. The size of this data set is 512 × 217 × 224, with a spatial resolution of 3.7 m. A total of 20 water absorption bands (108th–112th, 154th–167th, and 224th) were removed before the classification tasks. The data comprise 16 ground-truth classes, as shown in Table II. The false-color image and ground-truth map of the Salinas data set are shown in Fig. 6.



(a) (b)

Fig. 5. University of Pavia. (a) False-color image. (b) Ground-truth map.



(a) (b)

Fig. 6. Salinas. (a) False-color image. (b) Ground-truth map.

B. Experimental Configuration

All the handcrafted spatial–spectral feature-based classification experiments in this study are implemented in MATLAB 2014a on a computer with a 3.2-GHz Intel Core $i5$ processor and 8.0-GB memory with a Windows 10 operating system, and the deep-learning-based classification methods are implemented on the same computer equipped with a single 1060 Ti GPU.

In the following experiments, only five training samples are randomly selected per class for the Indian Pines and Salinas data sets, and only ten training samples are randomly selected per class for the University of Pavia data set, with the remaining samples used for testing. The numbers of training and testing samples for the three HSI data sets are shown in Table II. In this study, four indexes are used to evaluate the classification performance, i.e., overall accuracy (OA), average accuracy (AA), κ coefficient, and accuracy of each class. Because the training samples are randomly selected from the overall labeled pixels, the following classification results are all averaged over ten independent iterations.

C. Analysis of the Dimension l Reduced Using MNF

To reduce the computational complexity substantially and effectively remove the noise in the original HSI data sets, we apply MNF to the original HSIs to obtain the dimension-reduced HSI data sets. In this section, we present the results of several experiments to analyze the influence of the dimension l in the proposed SPMSFE method for classification performance. Fig. 7 shows the classification results in terms of the OA (%) and runtime (Sec.) of the

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TABLE II
NUMBERS OF TRAINING SAMPLES AND TESTING SAMPLES ON THE THREE HSI DATA SETS

Indian Pines				Pavia University				Salinas			
Class	Name	Train	Test	Class	Name	Train	Test	Class	Name	Train	Test
1	Alfalfa	5	41	1	Asphalt	10	6621	1	Broccoli-green-weeds 1	5	2004
2	Corn-no till	5	1423	2	Meadows	10	18639	2	Broccoli-green-weeds 2	5	3721
3	Corn-min till	5	825	3	Gravel	10	2089	3	Fallow	5	1971
4	Corn	5	232	4	Trees	10	3054	4	Fallow-rough-plow	5	1386
5	Grass/Pasture	5	478	5	Metrl sheet	10	1335	5	Fallow-smooth	5	2673
6	Grass/Trees	5	725	6	Grass/Trees	10	5019	6	Stubble	5	3954
7	Grass/Pasture-mowed	5	24	7	Bitumen	10	1320	7	Celery	5	3574
8	Hay-windrowed	5	473	8	Brick	10	3672	8	Grapes-untrained	5	11266
9	Oats	5	15	9	Shadows	10	937	9	Soil-vinyary-develop	5	6198
10	Soybeans-no till	5	987					10	Corn-senesced-green-weeds	5	3273
11	Soybeans-min till	5	2450					11	Lettuce-romaine-4wk	5	1063
12	Soybeans-clean till	5	588					12	Lettuce-romaine-5wk	5	1922
13	Wheat	5	200					13	Lettuce-romaine-6wk	5	911
14	Woods	5	1260					14	Lettuce-romaine-7wk	5	1065
15	Bldg-grass-tree-drives	5	381					15	Vineyard-untrained	5	7263
16	Stone-steel towers	5	88					16	Vineyard-vertical-trellis	5	1802
	Total	80	10169		Total	90	42686		Total	80	54049

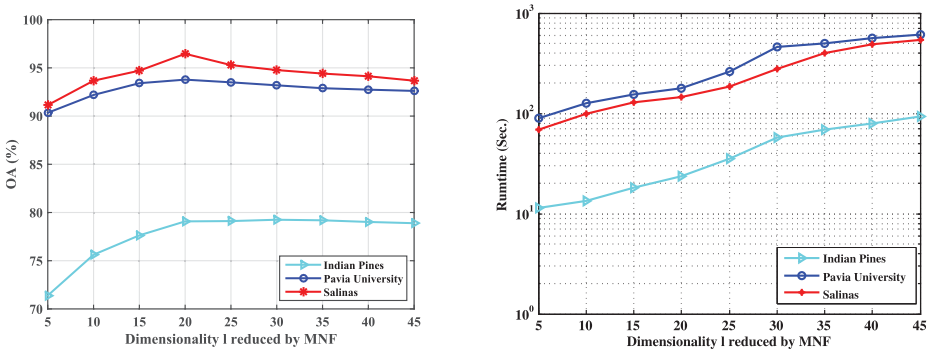


Fig. 7. Impacts of the dimension l in our proposed SPMSFE method for HSI classification. (a) OA(%) results. (b) Runtime (Sec.) results.

proposed SPMSFE when the dimension l is increased from 5 to 45 for all three HSI data sets.

From Fig. 7(a), we observe that the OA increases rapidly when the reduced dimension l ranges from 5 to 20 for all three data sets. As l increases, the OA shows a very slight increase for the Indian Pines data set and decreases to a certain extent for the University of Pavia and Salinas data sets. Since the MNF is applied to simultaneously retain the main information in the original data set while effectively removing noise, a relatively small dimension l prevents the MNF from retaining sufficient quantity of the main information in the original data set, whereas a relatively large dimension l would introduce some noise to reduce classification accuracy. In addition, we observe from Fig. 7(b) that the runtime always increases rapidly when the dimension l increases from 5 to 45 for all three test data sets. The main reason for this is that the dimensions of the mean feature, covariance descriptor, and the Gaussian feature would increase significantly with increasing l, thereby increasing the computational complexity considerably. Considering the classification accuracy and computational complexity together, we fix the dimension l of the proposed SPMSFE method as 20 in all subsequent experiments for all three data sets.

D. Analysis of Parameters L and M

In this section, the results of a series of experiments are presented to analyze the effects of two of the main parameters in the proposed method SPMSFE for HSI classification, i.e., the number of superpixels L and threshold of the superpixel-based neighbors M. We analyze the effects of these two parameters on the three test data sets by evaluating two indexes of classification accuracy, namely, the OA and kappa coefficient (κ).

For the Indian Pines and Salinas data sets, the number of superpixels L changes from 30 to 120 in steps of 10, while, for the University of Pavia data set, L increases from 150 to 350 in steps of 25. The threshold of the superpixel-based neighbors M ranges from 100 to 300 in steps of 20 for all three HSI data sets. The classification results in terms of the OA and κ for the proposed SPMSFE method for different L and M are shown in Fig. 8.

As observed from Fig. 8, the OA and κ both increase when the number of superpixels L increases from 20 to 60 for the Indian Pines and Salinas sets and from 150 to 250 for the University of Pavia set. However, the OA and κ decrease with a further increase in L; the main reason for this is that a small L introduces some dissimilar pixels, especially in the edge areas, while a large L cannot fully exploit the spatial information for classification. In addition, the optimal values of L for the Indian Pines, Salinas, and University of Pavia data sets are 60, 60, and 250, respectively. This is mainly attributed to the smoothing of large areas of the Indian Pines and Salinas sets with a simple geometric spatial structure, while the University of Pavia set contains a number of small and intricately detailed geometric structures. Hence,

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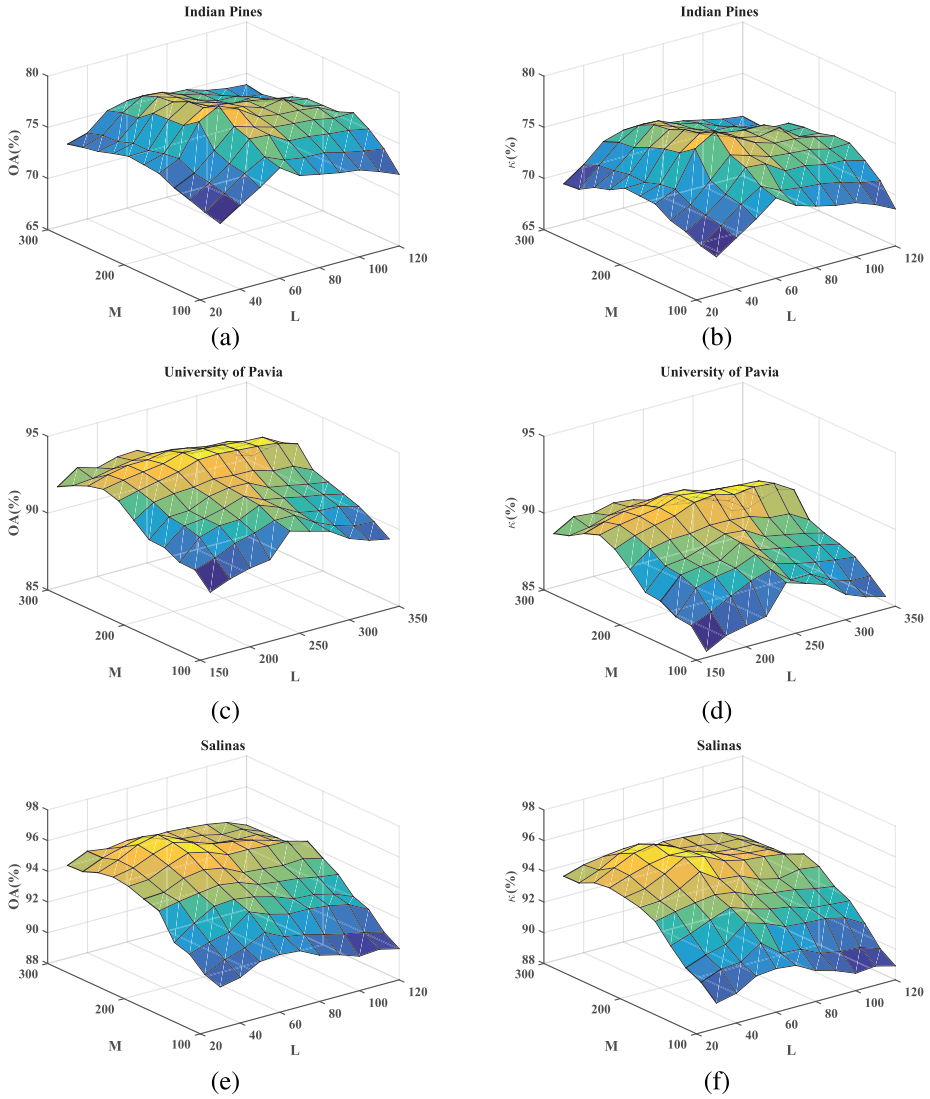


Fig. 8. Impacts of the number of superpixels L and the threshold of superpixel-based neighbors M in our proposed SPMSFE method for HSI classification. From top to bottom are the classification results for Indian Pines, Pavia University, and Salinas, where the left column represents the OA results and the right column denotes the κ results.

it is reasonable that a relatively small L is obtained for the Indian Pines and Salinas data sets, and a relatively large L is obtained for the University of Pavia data set. Therefore, the number of superpixels L in the proposed SPMSFE method is set to 60 for the Indian Pines and Salinas data sets and 250 for the University of Pavia data set.

From Fig. 8, we also note that the OA and κ both increase with the threshold of the superpixel-based neighbors M ranging from 100 to 180 (or 160) and decrease slightly with further increase in M for the Indian Pines set. A similar trend is also seen for the University of Pavia and Salinas sets. Due to the fact that the University of Pavia and Salinas data sets have larger sizes, the optimal M values are relatively large, i.e., 220–240 and 240–260, respectively. The main reason for this is that a small M cannot adequately utilize the spatial information within the superpixel-based neighbors, while a large M would incorporate dissimilar pixels. Since improved classification accuracies can be obtained for all three HSI data sets when M ranges from 180 to 240, we fix M as 220 in all subsequent experiments.

To summarize, all the parameters in our proposed method SPMSFE, including the dimension l reduced by MNF, the number of superpixels L, threshold of the superpixel-based neighbors M, sparsity level K, and minimum error limit ϵ, are set to the values indicated in Table III in the following experiments.

E. Analysis of Different Statistical Feature Extractions

This experiment was conducted on three widely used HSI data sets to demonstrate the advantages of the SPMSFE by comparing the proposed SPMSFE method with three FE methods by extracting any one kind of the multiple statistical features, including the mean feature extraction (Mean-FE) method, the covariance feature extraction (Cor-FE) method, and the Gaussian feature extraction (Gauss-FE) method.

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TABLE III

PARAMETERS FOR THE SUPERPIXEL-BASED MULTIPLE STATISTICAL FEATURE EXTRACTION

Parameters	Optimal values
l	20
L	60 (Indian Pines) 60 (Salinas) 250 (Pavia University)
M	220
K	3
ϵ	1×10^{-5}

TABLE IV

CLASSIFICATION RESULTS OF MEAN-FE, COR-FE, GAUSS-FE, AND SPMSFE METHODS FOR INDIAN PINES

Class	Mean-FE	Cor-FE	Gauss-FE	SPMSFE
1	**100** ± 0	99.27 ± 1.18	99.27 ± 1.65	**100** ± 0
2	71.00 ± 14.68	65.30 ± 18.94	76.08 ± 9.87	**76.40** ± 12.10
3	64.58 ± 10.41	**78.85** ± 13.16	70.72 ± 15.65	60.81 ± 15.05
4	92.20 ± 15.41	83.45 ± 14.59	82.20 ± 14.09	**92.24** ± 9.66
5	79.39 ± 10.81	79.37 ± 8.90	80.48 ± 2.33	83.03 ± 8.44
6	98.74 ± 1.89	88.55 ± 8.48	89.26 ± 6.49	**99.26** ± 0.60
7	**100** ± 0	**100** ± 0	**100** ± 0	**100** ± 0
8	99.58 ± 0	99.83 ± 0.09	**99.83** ± 0.09	99.58 ± 0
9	**100** ± 0	**100** ± 0	**100** ± 0	**100** ± 0
10	80.12 ± 11.37	76.48 ± 9.00	79.32 ± 13.23	**80.36** ± 6.23
11	56.52 ± 7.13	56.49 ± 13.16	59.13 ± 16.89	**63.83** ± 11.37
12	76.02 ± 9.32	71.92 ± 13.74	69.85 ± 11.40	**78.06** ± 15.31
13	99.50 ± 0	99.45 ± 0.60	**99.70** ± 0.35	99.50 ± 0
14	**92.56** ± 8.46	91.51 ± 5.99	87.17 ± 7.91	91.53 ± 8.22
15	87.90 ± 14.05	**94.07** ± 9.19	91.84 ± 9.26	87.45 ± 13.54
16	97.73 ± 0	97.95 ± 6.57	**100** ± 0	98.07 ± 0.77
OA	76.66 ± 4.04	75.63 ± 4.88	76.86 ± 4.21	**79.08** ± 2.93
AA	87.26 ± 2.85	86.42 ± 2.3)	86.61 ± 1.79	**88.13** ± 1.59
κ	73.77 ± 4.58	72.73 ± 5.34	74.02 ± 4.63	**76.54** ± 3.19

TABLE V

CLASSIFICATION RESULTS OF MEAN-FE, COR-FE, GAUSS-FE, AND SPMSFE METHODS FOR PAVIA UNIVERSITY

Class	Mean-FE	Cor-FE	Gauss-FE	SPMSFE
1	77.76 ± 9.06	74.91 ± 4.97	83.16 ± 8.63	**85.32** ± 8.02
2	88.36 ± 3.78	93.32 ± 2.72	94.57 ± 5.68	**94.87** ± 2.62
3	87.17 ± 6.91	**96.06** ± 2.82	91.63 ± 8.35	94.40 ± 3.89
4	82.81 ± 6.42	**97.58** ± 1.36	97.29 ± 1.59	96.30 ± 2.22
5	99.40 ± 0	**99.96** ± 0.12	99.23 ± 1.47	99.47 ± 0.19
6	94.35 ± 8.31	95.37 ± 4.10	96.26 ± 3.63	**97.53** ± 3.07
7	**99.76** ± 0.35	91.17 ± 8.64	96.34 ± 1.87	96.11 ± 1.57
8	84.46 ± 8.64	81.63 ± 6.81	92.13 ± 6.93	**94.21** ± 3.29
9	66.82 ± 5.59	79.36 ± 4.77	**94.25** ± 2.88	88.41 ± 3.87
OA	86.85 ± 2.00	89.97 ± 1.82	93.03 ± 3.61	**93.76** ± 1.93
AA	86.76 ± 1.76	89.93 ± 1.44	93.87 ± 2.36	**94.07** ± 1.43
κ	82.94 ± 2.49	86.92 ± 2.30	90.92 ± 4.55	**91.84** ± 2.50

TABLE VI

CLASSIFICATION RESULTS OF MEAN-FE, COR-FE, GAUSS-FE, AND SPMSFE METHODS FOR SALINAS

Class	Mean-FE	Cor-FE	Gauss-FE	SPMSFE
1	98.89 ± 1.20	99.93 ± 0.13	99.95 ± 0.08	**99.97** ± 0.04
2	**100** ± 0	98.79 ± 1.84	99.69 ± 0.94	99.88 ± 0.39
3	**100** ± 0	96.42 ± 6.49	99.51 ± 1.10	**99.94** ± 0.12
4	94.77 ± 5.15	97.63 ± 3.83	99.15 ± 0.55	**99.45** ± 0.41
5	**98.75** ± 0.33	96.06 ± 1.77	98.10 ± 0.80	98.18 ± 0.32
6	99.82 ± 0	99.82 ± 0.10	**99.90** ± 0.04	99.86 ± 0.04
7	**99.92** ± 0	99.32 ± 0.75	99.80 ± 0.11	99.80 ± 0.11
8	79.90 ± 10.28	83.72 ± 10.89	86.91 ± 5.12	**88.06** ± 4.51
9	**99.99** ± 0.02	99.14 ± 1.13	99.99 ± 0.01	99.95 ± 0.09
10	89.95 ± 9.23	**92.46** ± 5.02	88.74 ± 9.85	90.96 ± 11.15
11	96.07 ± 3.84	**97.87** ± 1.89	96.22 ± 2.93	96.81 ± 2.75
12	**99.48** ± 0	87.49 ± 14.33	94.88 ± 6.00	94.66 ± 8.87
13	**97.83** ± 3.79	93.02 ± 5.46	96.16 ± 1.70	96.66 ± 2.95
14	89.84 ± 5.21	**95.11** ± 2.58	92.90 ± 7.68	93.01 ± 3.18
15	86.13 ± 8.41	93.28 ± 6.15	88.20 ± 9.50	**93.42** ± 3.74
16	96.52 ± 0.69	99.44 ± 0.41	99.50 ± 0.45	**99.53** ± 0.47
OA	92.63 ± 1.46	93.94 ± 2.18	94.36 ± 1.44	**95.47** ± 1.27
AA	95.49 ± 0.72	95.61 ± 1.44	96.23 ± 0.97	**96.88** ± 0.72
κ	91.82 ± 1.60	93.31 ± 2.40	93.71 ± 1.60	**94.96** ± 1.41

Here, the identification of the superpixel-based neighbors and the modeling of the classifier and some of the parameters are the same for the four different FE methods. As it is not reasonable to use the same L and M for comparison of the Cor-FE, Mean-FE, and Gauss-FE, we performed a mass of experiments with the L and M changing to evaluate the classification accuracy and obtained the optimal values for the Cor-FE, Mean-FE, and Gauss-FE. The parameters L and M in the Cor-FE, Mean-FE, and Gauss-FE are set as $\{[60, 200], [60, 160], [60, 240]\}$ for Indian Pines, $\{[175, 225], [200, 175], [300, 175]\}$ for University of Pavia, and $\{[60, 260], [80, 180], [60, 280]\}$ for Salinas. There are four indexes to evaluate the classification performance, i.e., OA, AA, the κ coefficient, and the accuracy of each class. The mean classification results with standard variance (in percentage) based on these four FE methods for Indian Pines, University of Pavia, and Salinas are, respectively, listed in Tables IV–VI.

Tables IV–VI show that our proposed SPMSFE achieved the best classification performance with the highest OA, AA, and κ coefficient for all the three HSI data sets. Moreover, all four FE methods are effective in improving the classification accuracy to some extent. The first main reason is that the framework of identifying superpixel-based neighbors can explore the spatial–spectral information accurately, which can improve the classification accuracy significantly. The second main reason is that all three kinds of statistical features are useful to exploit the spatial–spectral information from different perspectives for classification. In addition, the classification accuracy obtained by the Gauss-FE method was higher than that obtained by the Mean-FE and Cor-FE methods. This is mainly because the Gaussian model can estimate the mean vector and CM, which offers a way to simultaneously represent the characteristics of the mean feature and covariance feature. Moreover, the classification accuracies of some classes obtained by the Mean-FE (such as the 14th class of Indian Pines, the seventh class of Pavia University; and the second, third, fifth, seventh, ninth, 12th, and 13th classes of Salinas), and Cor-FE (such as the third and 15th classes of Indian Pines; third, fourth, and fifth classes of Pavia University; and 10th, 11th, and 14th classes of Salinas) are higher than that obtained by the Gauss-FE method. The major reason is that some regions belonging to those classes may not follow the Gaussian distribution, resulting in relatively lower classification accuracy. Although the Gaussian distribution cannot always be obtained in some areas of the HSI data sets, the mean feature and covariance descriptor can be chosen to better characterize each pixel in the proposed SPMSFE method. Because the three kinds of statistical features are complementary to each other, the proposed SPMSFE designs multiple kernels to map them into a uniform Hilbert space and construct a multiple-task SRC model to effectively fuse them to further improve the classification accuracy.

F. Analysis of Different Classifiers

This experiment was conducted to demonstrate the effectiveness of the proposed SPMSFE method for HSI classification by embedding the multiple statistical features into

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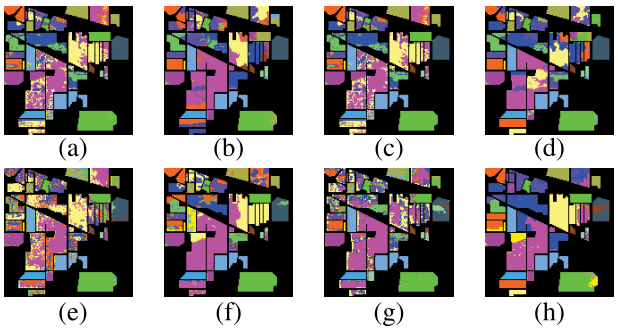


Fig. 9. Classification maps obtained by the state-of-the-art classification methods for Indian Pines. (a) SSHGDA. (b) SFMKSRC. (c) LCMR. (d) SRST. (e) ISSR-DCNN. (f) MCMs-2DCNN. (g) S-DMM. (h) Proposed SPMSFE.

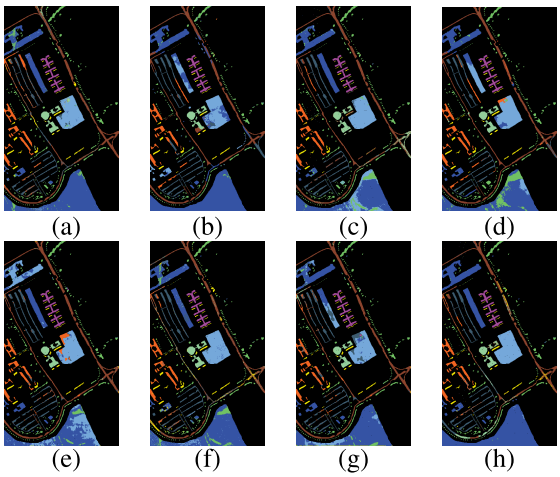


Fig. 10. Classification maps obtained by the state-of-the-art classification methods for Pavia University. (a) SSHGDA. (b) SFMKSRC. (c) LCMR. (d) SRST. (e) ISSR-DCNN. (f) MCMs-2DCNN. (g) S-DMM. (h) Proposed SPMSFE.

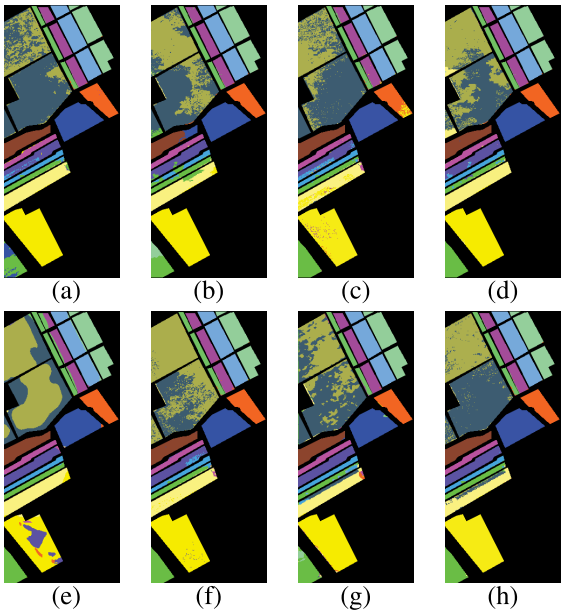


Fig. 11. Classification maps obtained by the state-of-the-art classification methods for Salinas. (a) SSHGDA. (b) SFMKSRC. (c) LCMR. (d) SRST. (e) ISSR-DCNN. (f) MCMs-2DCNN. (g) S-DMM. (h) Proposed SPMSFE.

several different classical classifiers, including the K-nearest neighbors (KNN) classifier (SPMSF-KNN), SVM classifier (SPMSF-SVM), and SR classifier (i.e., the proposed method SPMSFE). Before classification based on KNN and SVM, the multiple statistical features are mapped into a uniform Hilbert space by combing multiple kernels. Moreover, the classical classifiers with the original spectral feature are utilized for comparison to further demonstrate the significant improvement obtained by our proposed SPMSFE. Here, the parameter K in KNN and SPMSF-KNN is fixed as 1, and the parameters in SVM and SPMSF-SVM are determined by five-fold cross-validation. The mean classification results with standard variance (in percentage) on the three widely used HSI data sets based on KNN, SVM, SRC, SPMSF-KNN, SPMSF-SVM, and SPMSFE are, respectively, shown in Tables VII–IX.

From Tables VII–IX, it can be observed that the classification accuracies of SPMSF-KNN, SPMSF-SVM, and SPMSFE are significantly higher than those of the corresponding KNN, SVM, and SRC, which adequately confirms that the multiple statistical features extracted by the proposed SPMSFE are effective to explore spatial–spectral information and significantly improve the classification accuracy. In addition, the proposed SPMSFE with the MTKSRC classifier outperforms the SPMSFE with the KNN and SVM classifiers, which further demonstrates the superiority of the SPMSFE with the MTKSRC classification framework.

G. Comparison to the State-of-the-Art Algorithms

This experiment is performed to verify the superiorities of the proposed SPMSFE method by comparing it to various state-of-the-art spatial–spectral classification methods, such as the spatial–spectral hypergraph discriminant analysis (SSHGDA)-based feature learning method [66], the superpixel-feature-based multiple kernel sparse representation classification (SFMKSRC) method [67], the local covariance matrix representation (LCMR) [32], the semisupervised classification via spatial-regulated self-training (SRST) [68], the intraclass similarity structure representation and deep convolutional neural network-based classification (ISSR-DCNN) method [69], the MCM-based 2-D convolutional neural network classifier (MCMs-2DCNN) [31], and the similarity-based deep metric model (S-DMM)-based classification method [70]. The SSHGDA, SFMKSRC, and LCMR methods are the latest and state-of-the-art handcraft FE-based classification methods, whereas the SRST, ISSR-DCNN, MCMs-2DCNN, and S-DMM methods are the latest and state-of-the-art deep learning-based classification methods. Moreover, the SSHGDA, LCMR, SRST, ISSR-DCNN, and S-DMM methods are specially developed for HSI classification when the training samples are very limited, where SRST belongs to semisupervised classification method. Similar to the studies from [32], [66]–[68], [31], [69], and [70], the parameters of the compared SSHGDA, SFMKSRC, LCMR, SRST, ISSR-DCNN, MCMs-2DCNN, and S-DMM are set to their optimal values. The numbers of the training and testing samples in the above mentioned state-

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TABLE VII

CLASSIFICATION RESULTS OF KNN, SVM, KSRC, SPMSF-KNN, SPMSF-SVM, AND THE PROPOSED SPMSFE FOR INDIAN PINES

Class	KNN	SVM	KSRC	SPMSF-KNN	SPMSF-SVM	SPMSFE
1	82.44 ± 7.95	88.05 ± 6.65	85.37 ± 7.71	96.34 ± 2.07	97.80 ± 1.80	**100** ± 0
2	46.09 ± 7.05	61.12 ± 8.49	44.31 ± 11.05	63.93 ± 10.63	**77.12** ± 7.54	76.40 ± 12.10
3	56.32 ± 8.41	56.82 ± 7.32	48.56 ± 6.87	71.32 ± 9.52	**80.00** ± 11.90	60.81 ± 15.05
4	59.87 ± 10.91	65.09 ± 10.45	52.59 ± 14.90	80.39 ± 21.20	88.32 ± 12.40	**92.24** ± 9.66
5	74.62 ± 5.74	78.83 ± 20.47	77.68 ± 11.25	75.15 ± 9.31	79.27 ± 10.32	**83.03** ± (8.44
6	70.44 ± 11.28	88.29 ± 6.53	92.97 ± 3.83	69.19 ± 12.10	84.94 ± 12.43	**99.26** ± 0.60
7	93.48 ± 11.64	98.26 ± 2.25	96.52 ± 1.83	100 ± 0	**100** ± 0	**100** ± 0
8	65.71 ± 14.32	86.22 ± 5.97	88.88 ± 10.06	99.79 ± 0	**99.79** ± 0	99.58 ± 0
9	99.33 ± 2.11	**100** ± 0	96.67 ± 7.20	95.33 ± 6.32	**100** ± 0	**100** ± 0
10	52.18 ± 12.84	60.83 ± 11.54	59.58 ± 9.83	74.93 ± 9.15	**81.76** ± 5.68	80.36 ± 6.23
11	45.82 ± 8.49	40.72 ± 12.45	44.82 ± 10.58	58.61 ± 7.67	52.10 ± 7.90	**63.83** ± 11.37
12	48.91 ± 11.91	64.44 ± 4.98	52.19 ± 10.85	69.88 ± 16.95	75.54 ± 13.27	**78.06** ± 5.31
13	97.40 ± 2.07	99.30 ± 0.35	97.65 ± 1.03	99.45 ± 0.16	**99.95** ± 0.16	99.50 ± 0
14	77.74 ± 10.45	79.07 ± 11.26	78.14 ± 9.55	**94.83** ± 5.252	91.09 ± 8.26	91.53 ± 8.22
15	50.00 ± 8.24	67.01 ± 16.89	52.07 ± 13.87	64.15 ± 16.32	87.35 ± 14.87	**87.45** ± 13.54
16	86.82 ± 3.87	90.34 ± 4.68	89.43 ± 4.05	98.18 ± 2.09	**98.75** ± 1.96	98.07 ± 0.77
OA	57.66 ± 2.96	63.75 ± 3.00	60.27 ± 2.89	72.67 ± 2.81	76.57 ± 2.78	**79.08** ± 2.93
AA	69.20 ± 1.32	76.52 ± 1.54	72.34 ± 1.93	81.97 ± 2.47	87.11 ± 2.44	**88.13** ± 1.59
κ	52.57 ± 3.23	59.55 ± 3.08	55.51 ± 3.31	69.32 ± 3.07	73.79 ± 3.05	**76.54** (3.19)

TABLE VIII

CLASSIFICATION RESULTS OF KNN, SVM, KSRC, SPMSF-KNN, SPMSF-SVM, AND THE PROPOSED SPMSFE FOR PAVIA UNIVERSITY

Class	KNN	SVM	KSRC	SPMSF-KNN	SPMSF-SVM	SPMSFE
1	63.76 ± 6.89	66.98 ± 8.18)	48.35 ± 7.97	76.06 ± 4.65	82.20 ± 6.99	**85.32** ± 8.02
2	64.67 ± 7.59	76.70 ± 7.54	61.44 ± 8.20	77.53 ± 10.25	87.80 ± 10.58	**94.87** ± 2.62
3	62.82 ± 8.05	69.12 ± 10.65	63.90 ± 9.49	85.59 ± 5.71	**95.13** ± 4.96	94.40 ± 3.89
4	86.35 ± 4.90	84.02 ± 9.18	80.76 ± 2.26	86.96 ± 3.70	**97.02** ± 2.58	96.30 ± 2.22
5	**99.86** ± 0.17	99.81 ± 0.04	99.53 ± 0.35	99.40 ± 0	99.26 ± 0.40	99.47 ± 0.19
6	76.88 ± 9.11	79.01 ± 11.94	66.12 ± 7.82	93.74 ± 7.51	94.78 ± 4.67	**97.53** ± 3.07
7	84.79 ± 3.81	78.30 ± 12.45	78.51 ± 9.65	95.23 ± 6.34	**96.95** ± 2.29	96.11 ± 1.57
8	72.28 ± 11.09	73.37 ± 8.73	44.97 ± 8.62	83.42 ± 9.86	91.56 ± 4.98	**94.21** ± 3.29
9	95.35 ± 2.17	99.68 ± 0.19	95.23 ± 2.07	**99.75** ± 0.48	86.25 ± 7.27	88.41 ± 3.87
OA	70.48 ± 3.43	76.60 ± 3.54	65.63 ± 3.56	82.51 ± 4.25	89.70 ± 4.59	**93.76** ± 1.93
AA	78.43 ± 2.12	80.78 ± 2.01	72.20 ± 1.19	89.35 ± 1.53	92.33 ± 1.40	**94.07** ± 1.43
κ	66.41 ± 3.77)	70.17 ± 4.13	55.49 ± 3.43	75.02 ± 4.84	86.78 ± 5.59	**91.84** ± 2.50

TABLE IX

CLASSIFICATION RESULTS OF KNN, SVM, KSRC, SPMSF-KNN, SPMSF-SVM, AND THE PROPOSED SPMSFE FOR SALINAS

Class	KNN	SVM	KSRC	SPMSF-KNN	SPMSF-SVM	SPMSFE
1	97.84 ± 1.65	98.90 ± 1.61	98.63 ± 1.20	99.90 ± 0.15	99.60 ± 1.18	**99.97** ± 0.04
2	96.94 ± 0.91	98.29 ± 1.10	98.81 ± 0.92	97.24 ± 0.31	**100** ± 0	99.88 ± 0.39
3	84.91 ± 9.59	94.34 ± 8.01	98.78 ± 1.69	95.80 ± 8.21	**100** ± 0	99.94 ± 0.12
4	95.23 ± 2.53	97.89 ± 2.51	96.08 ± 3.63	96.23 ± 5.59	98.56 ± 0.46	**99.45** ± 0.41
5	95.65 ± 3.50	96.28 ± 4.79	97.68 ± 0.52	**98.81** ± 0.93	98.65 ± 1.25	98.18 ± 0.32
6	99.59 ± 0.19	99.58 ± 0.18	99.56 ± 0.22	99.82 ± 0	99.82 ± 0	**99.86** ± 0.04
7	94.91 ± 15.57	98.79 ± 2.83	99.82 ± 0.06	**99.89** ± 0.05	99.89 ± 0	99.80 ± (0.11
8	55.62 ± 15.73	61.72 ± 12.04	67.09 ± 13.18	72.23 ± 9.87	71.90 ± 12.16	**88.06** ± 4.51
9	98.84 ± 0.53	99.60 ± (0.57	99.22 ± 0.79	99.73 ± 0.86	**99.98** ± 0.03	99.95 ± 0.09
10	75.41 ± 13.08	81.48 ± 9.21	86.12 ± 6.58	88.70 ± 8.82	86.99 ± 7.17	**90.96** ± 11.15
11	92.06 ± 2.69	95.31 ± 3.70	93.99 ± 2.48	93.76 ± 3.10	94.59 ± 4.15	**96.81** ± 2.75
12	93.41 ± 10.76	95.55 ± 6.59	**99.28** ± 0.69	99.07 ± 1.97	95.43 ± 4.44	94.66 ± 8.87
13	98.64 ± 0.52	**99.06** ± 0.61	96.95 ± 5.13	94.97 ± 3.57	88.09 ± 16.24	96.66 ± 2.95
14	93.45 ± 2.33	92.03 ± 4.55	88.13 ± 3.19	92.70 ± 5.34	**94.06** ± 0.99	93.01 ± 3.18
15	62.71 ± 10.96	67.87 ± 9.60	70.75 ± 8.53	73.84 ± 10.37	88.57 ± 5.70	**93.42** ± 3.74
16	87.95 ± 7.19	95.67 ± 2.670	96.30 ± 1.47	87.21 ± 8.69	98.27 ± 1.01	**99.53** ± 0.47
OA	81.63 ± 3.03	85.25 ± 2.39	87.29 ± 2.25	88.65 ± 1.28	91.03 ± 2.70	**95.47** ± 1.27
AA	88.95 ± 1.45	92.02 ± 1.42	92.95 ± 0.84	93.12 ± 1.11	94.65 ± 1.31	**96.88** ± 0.72
κ	79.62 ± 3.30	83.63 ± 2.63	85.89 ± 2.47)	87.38 ± 1.41	90.05 ± 2.97	**94.96** ± 1.41

of-the-art classification methods are set to the same values, as shown in Table I. The mean classification results with standard variance (in percentage) of the aforementioned eight methods for Indian Pines, University of Pavia, and Salinas are, respectively, reported in Tables X–XII. In addition, the classification maps obtained by these classification methods are displayed in Figs. 9–11, respectively.

It can be seen from Tables X–XII that the classification accuracy achieved by the proposed SPMSFE method is obviously superior to that of the compared state-of-the-art methods by a large margin (more than 3% higher OA value compared to the other seven methods). The main reason for this observation is that the multiple statistical features extracted by the SPMSFE, i.e., the mean feature, the covariance descriptor, and the Gaussian feature, can completely explore the spatial geometry information, the tight correlations between different spectral bands, and the spatial–spectral variations, respectively, to describe the pixels from different perspectives. Moreover, due to the fact that the three kinds of statistical features are complementary to each other, we designed multiple kernels to map them into a uniform Hilbert space and construct an MTKSRC model to effectively fuse them for classification. The MTKSRC model based on multiple statistical features can offer a significant improvement in classification accuracy with more robustness, particularly when the training samples are limited. Although the compared deep learning-based classification methods, i.e., SRST, ISSR-DCNN, MCMs-2DCNN, and S-DMM, automatically extract both the simple and complicated spectral-spatial features to improve the classification accuracy remarkably, they require a relatively large number of samples to train the deep networks. Therefore, the proposed SPMSFE method is distinctly superior to the state-of-the-art deep learning-based classification methods when the training samples are limited. The better classification performance of the proposed SPMSFE method can be also illustrated from the classification maps displayed in Figs. 9–11.

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TABLE X

CLASSIFICATION RESULTS OF SSHGDA, SFMKSRC, LCMR, SRST, ISSR-DCNN, MCMS-2DCNN, S-DMM, AND THE PROPOSED SPMSFE METHODS FOR INDIAN PINES

Class	SSHGDA [66]	SFMKSRC [67]	LCMR [32]	SRST [68]	ISSR-DCNN [69]	MCMs-2DCNN [31]	S-DMM [70]	SPMSFE
1	**100** ± 0	98.35 ± 1.23	99.76 ± 0.77	**100** ± 0	68.29 ± 2.75	95.24 ± 4.11	92.44 ± 7.84	**100** ± 0
2	62.04 ± 9.47	58.41 ± 7.49	63.33 ± 17.38	67.50 ± 6.39	42.16 ± 13.27	54.56 ± 17.37	33.11 ± 13.66	**76.40** ± 12.10
3	56.02 ± 15.29	55.71 ± 13.63	51.68 ± 13.57	**80.48** ± 7.15	41.82 ± 15.91	54.00 ± 21.35	38.11 ± 16.52	60.81 ± 15.05
4	92.21 ± 7.89	86.74 ± 9.42	87.93 ± 11.87	**97.84** ± 2.37	76.72 ± 11.28	94.57 ± 3.79	61.72 ± 25.73	92.24 ± 9.66
5	77.05 ± 11.95	73.28 ± 11.49	79.02 ± 20.13	80.15 ± 5.35	**87.03** ± 5.49	63.67 ± 13.29	72.13 ± 13.00	83.03 ± 8.44
6	88.26 ± 11.25	89.37 ± 5.92	83.60 ± 8.14	78.34 ± 15.81	88.55 ± 7.23	85.40 ± 7.93	87.89 ± 6.36	**99.26** ± 0.60
7	**100** ± 0	99.15 ± 1.58	**100** ± 0	**100** ± 0	95.96 ± 2.03	87.50 ± 19.36	100 ± 0	**100** ± 0
8	98.52 ± 1.27	**100** ± 0	98.84 ± 2.86	99.79 ± 0	95.14 ± 2.75	99.79 ± 0.04	84.65 ± 7.68	99.58 ± 0
9	**100** ± 0	**100** ± 0	**100** ± 0	**100** ± 0	**100** ± 0	**100** ± 0	**100** ± 0	**100** ± 0
10	73.11 ± 25.38	69.39 ± 10.49	69.45 ± 11.98	73.73 ± 9.41	51.29 ± 12.93	77.48 ± 21.35	58.77 ± 10.92	**80.36** ± 6.23
11	60.88 ± 13.72	60.18 ± 9.50	59.88 ± 12.92	61.35 ± 13.85	38.45 ± 15.79	49.23 ± 23.79	52.67 ± 19.68	**63.83** ± 11.37
12	72.63 ± 17.93	62.35 ± 12.04	71.14 ± 17.44	74.97 ± 11.73	73.81 ± 17.42	71.34 ± 19.81	45.07 ± 13.21	**78.06** ± 15.31
13	98.99 ± 2.37	97.35 ± 10.65	99.15 ± 0.63	99.15 ± 0.06	98.50 ± 0.85	99.50 ± 0	98.95 ± 1.38	**99.50** ± 0
14	89.92 ± 7.61	80.59 ± 6.49	95.27 ± 5.93	88.49 ± 10.09	88.49 ± 7.49	99.60 ± 0.15	84.25 ± 11.26	**91.53** ± 8.22
15	84.74 ± 10.93	81.22 ± 11.48	79.74 ± 10.37	**98.43** ± 0.35	38.85 ± 12.37	85.86 ± 15.72	59.71(±15.75	87.45 ± 13.54
16	96.55 ± 3.17	98.25 ± 0.56	96.59 ± 3.71	97.50 ± 2.35	84.09 ± 6.41	94.38 ± 15.75	**98.64** ± 1.50	98.07 ± 0.77
OA	73.07 ± 4.57	70.42 ± 2.84)	72.83 ± 3.64	76.14 ± 3.27	60.01 ± 4.77	70.33 ± 3.57	59.83 ± 3.65	**79.08** ± 2.93
AA	84.43 ± 4.11	81.90 ± 2.31	83.46 ± 3.15	87.12 ± 2.07	73.05 ± 4.53	83.24 ± 3.02	73.01 ± 1.52	**88.13** ± 1.59
κ	69.97 ± 4.83	66.48 ± 3.11	69.43 ± 4.04	73.67 ± 3.71	55.26 ± 5.03	67.02 ± 3.25	55.06 ± 3.53	**76.54** ± 3.19

TABLE XI

CLASSIFICATION RESULTS OF SSHGDA, SFMKSRC, LCMR, SRST, ISSR-DCNN, MCMS-2DCNN, S-DMM, AND THE PROPOSED SPMSFE METHODS FOR PAVIA UNIVERSITY

Class	SSHGDA [66]	SFMKSRC [67]	LCMR [32]	SRST [68]	ISSR-DCNN [69]	MCMs-2DCNN [31]	S-DMM [70]	SPMSFE
1	84.33 ± 4.73	88.03 ± 4.36	82.36 ± 6.36	64.93 ± 9.21	69.08 ± 7.42	89.64 ± 8.35	**93.44** ± 4.10	85.32 ± 8.02
2	89.03 ± 6.85	79.89 ± 9.47	86.13 ± 4.82	85.33 ± 5.73	65.17 ± 11.61	84.37 ± 13.29	79.51 ± 11.47	**94.87** ± 2.62
3	90.87 ± 11.29	89.22 ± 8.63	87.73 ± 7.92	85.43 ± 7.15	73.75 ± 12.15	**96.22** ± 3.72	89.82 ± 4.91	94.40 ± 3.89
4	**98.27** ± 2.45	92.37 ± 6.11	96.32 ± 3.08	90.65 ± 3.79	86.11 ± 9.47	91.52 ± 7.39	94.39 ± 3.40	96.30 ± 2.22
5	99.85 ± 0.21	99.90 ± 0.15	99.10 ± 2.03	99.93 ± 0.05	98.88 ± 0.35	99.70 ± 0.21	**99.99** ± 0.02	99.47 ± 0.19
6	97.51 ± 2.47	87.25 ± 6.39	91.08 ± 5.65	88.47 ± 8.39	94.30 ± 4.39	90.67 ± 7.87	83.97 ± 10.32	**97.53** ± 3.07
7	72.21 ± 23.36	94.18 ± 8.57)	95.60 ± 3.09	**99.09** ± 0.75	90.61 ± 9.77	96.13 ± 2.37	97.95 ± 1.53	96.11 ± 1.57
8	76.97 ± 15.72	88.79 ± 4.38	83.19 ± 8.88	88.55 ± 5.63	82.21 ± 7.34	72.68 ± 25.35	91.39 ± 4.99	**94.21** ± 3.29
9	92.33 ± 7.95	93.56 ± 5.31	94.91 ± 3.04	78.66 ± 13.46	87.51 ± 7.11	89.64 ± 13.77	**99.97** ± 0.05	88.41 ± 3.87
OA	88.93 ± 1.59	85.50 ± 3.05	87.58 ± 2.20	83.99 ± 3.95	74.92 ± 4.53	86.97 ± 3.79	86.45 ± 4.37	**93.76** ± 1.93)
AA	89.03 ± 2.49	90.35 ± 2.15	90.71 ± 1.44	86.78 ± 2.43	83.07 ± 3.69	90.06 ± 3.13	92.27 ± 1.27	**94.07** ± 1.43
κ	85.63 ± 2.07	83.07 ± 3.79	83.96 ± 2.74	80.07 ± 3.25	68.84 ± 4.37	83.18 ± 4.11	82.72 ± 5.19	**91.84** ± 2.50

TABLE XII

CLASSIFICATION RESULTS OF SSHGDA, SFMKSRC, LCMR, SRST, ISSR-DCNN, MCMS-2DCNN, S-DMM, AND THE PROPOSED SPMSFE FOR SALINAS

Class	SSHGDA [66]	SFMKSRC [67]	LCMR [32]	SRST [68]	ISSR-DCNN [69]	MCMs-2DCNN [31]	S-DMM [70]	SPMSFE
1	99.95 ± 0.05	98.90 ± 4.27	99.97 ± 0.06	99.90 ± 0.09	99.50 ± 0.12	99.95 ± 0.09	98.96 ± 2.09	**99.97** ± 0.04
2	94.46 ± 21.37	99.39 ± 1.59	75.72 ± 10.41	**100** ± 0	97.53 ± 3.71	61.25 ± 25.69	98.23 ± 2.93	99.88 ± 0.39
3	99.09 ± 0.59	99.55 ± 1.75	97.98 ± 3.24	93.30 ± 4.75	95.28 ± 3.16	99.34 ± 0.35	94.04 ± 7.62	**99.94** ± 0.12
4	95.00 ± 3.7	95.87 ± 5.77	96.18 ± 11.76	98.27 ± 1.13	96.40 ± 4.97	99.35 ± 0.57	99.17 ± 1.25	**99.45** ± 0.41
5	93.61 ± 4.75	94.11 ± 9.36	90.66 ± 18.51	98.08 ± 0.61	95.85 ± 7.63	95.44 ± 3.95	97.15 ± 3.61	**98.18** ± 0.32
6	99.77 ± 1.13	99.75 ± 0	99.22 ± 0.71	99.22 ± 0.78	99.82 ± 0.08	98.23 ± 1.21	99.38 ± 1.16	**99.86** ± 0.04
7	97.09 ± 2.51	99.58 ± 1.33	96.16 ± 4.71	**99.92** ± 0.03	50.59 ± 21.73	98.77 ± 2.83	99.22 ± 1.96	99.80 ± 0.11
8	79.50 ± 17.49	60.04 ± 12.59	84.35 ± 7.31	59.80 ± 13.75	29.04 ± 13.57	78.28 ± 23.29	67.35 ± 10.61	**88.06** ± 4.51
9	82.99 ± 17.75	99.87 ± 0.24	83.62 ± 13.71	99.95 ± 0	98.79 ± 1.12	94.27 ± 5.73	97.98 ± 2.45	**99.95** ± 0.09
10	94.60 ± 5.43	90.31 ± 6.83	91.23 ± 9.55)	**95.08** ± 3.11	80.45 ± 11.35	94.33 ± 3.85	83.44 ± 10.60	90.96 ± 11.15
11	94.97 ± 3.25	98.13 ± 1.41	**99.97** ± 0.05	95.48 ± 2.73	87.11 ± 9.48	95.05 ± 4.93	96.12 ± 4.01	96.81 ± 2.75
12	84.84 ± 9.73	92.49 ± 9.52	96.75 ± 5.65	97.50 ± 1.35	**99.17** ± 1.89	84.04 ± 11.73	97.74 ± 6.34	94.66 ± 8.87
13	93.13 ± 2.77	94.37 ± 3.69	97.46 ± 3.02	95.28 ± 3.74	98.24 ± 2.03	99.01 ± 1.05	**99.66** ± 0.35	96.66 ± 2.95
14	**96.61** ± 3.37	93.41 ± 3.27	95.75 ± 3.80	94.93 ± 2.95	92.58 ± 5.75	94.09 ± 2.93	94.37 ± 6.10	93.01 ± 3.18
15	92.41 ± 7.39	80.83 ± 11.63	83.80 ± 8.45	**99.55** ± 0.35	85.03 ± 10.57	85.96 ± 9.81	77.73 ± 7.37	93.42 ± 3.74
16	82.61 ± 15.45	89.42 ± 7.59	92.81 ± 8.39	99.00 ± 0.79	90.34 ± 9.59	75.10 ± 23.75	91.88 ± 6.11	**99.53** ± 0.47
OA	89.94 ± 2.77	87.05 ± 2.01)	89.05 ± 3.03	90.46 ± 3.37	77.16 ± 5.73	87.75 ± 5.32	87.79 ± 2.74	**95.47** ± 1.27
AA	92.54 ± 1.89	92.87 ± 1.15	92.60 ± 2.53	95.27 ± 1.21	87.23 ± 3.74	90.78 ± 4.63	93.28 ± 1.44	**96.88** ± 0.72)
κ	88.85 ± 2.83	85.87 ± 2.03	87.84 ± 3.35	89.67 ± 2.99	74.82 ± 5.11	86.39 ± 5.57	86.45 ± 3.02	**94.96** ± 1.41

TABLE XIII

COMPUTATIONAL TIME (SEC.) OF SSHGDA, SFMKSRC, LCMR, SRST, ISSP-DCNN, MCMS-2DCNN, S-DMM, AND THE PROPOSED SPMSFE ON INDIAN PINES, PAVIA UNIVERSITY, AND SALINAS

Test image	SSHGDA	SFMKSRC	LCMR	SRST	ISSP-DCNN	MCMs-2DCNN	S-DMM	SPMSFE
Indian Pines	38.92	42.37	12.25	1347.95	1848.22	643.61	541.72	21.37
Pavia University	329.88	121.49	117.12	1681.35	2118.46	2845.37	2169.69	177.89
Salinas	252.61	366.72	186.58	1571.61	2312.57	2471.95	2492.38	146.26

H. Analysis of the Number of Training Samples

This experiment is conducted to report the effect of the number of training samples to further highlight the advantages of our proposed SPMSFE method. The classification results in terms of OA with the training number per class increasing from 3 to 10 for Indian Pines and Salinas and from 5 to 12 for the University of Pavia are, respectively, shown in Fig. 12(a)–(c). As can be observed from Fig. 12, the proposed SPMSFE method achieved the best classification accuracy with different training sample numbers for all the three HSI data sets. The proposed SPMSFE is clearly superior, particularly when the training samples are very limited.

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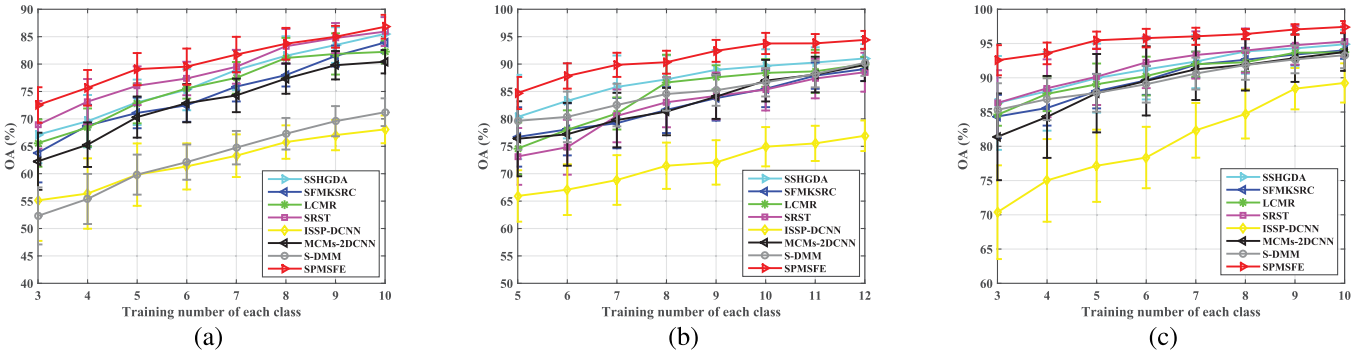


Fig. 12. Classification performance of the three HSI data sets with the training number of each class increasing from 3 to 10 (or from 5 to 12) by SSHGDA, SFMKSRC, LCMR, SRST, ISSP-DCNN, MCMs-2DCNN, S-DMM, and the proposed SPMSFE. (a) Indian Pines. (b) Pavia University. (c) Salinas.

I. Analysis of the Computational Speed

Finally, we analyze the computational speed of the proposed SPMSFE method by comparing it with the aforementioned state-of-the-art classification methods. The computational times of SSHGDA, SFMKSRC, LCMR, SRST, ISSP-DCNN, MCMs-2DCNN, S-DMM, and SPMSFE for Indian Pines, University of Pavia, and Salinas are listed in Table XIII. From Table XIII, we can observe that the computational speed of the proposed SPMSFE is faster than that of SSHGDA, SFMKSRC, SRST, ISSP-DCNN, MCMs-2DCNN, and S-DMM but slower than that of LCMR for all the three HSI data sets. Although multiple statistical FE leads to high computational costs, the dimensional reduction by MNF can reduce the computational complexity significantly. As a large number of parameters in the deep networks need to be tuned, the computational speed of the SPMSFE is much faster than that of the deep-learning-based classification methods. In addition, the SPMSFE improves the classification accuracy by a large margin compared to the other methods. Therefore, the classification performance of the SPMSFE considering the accuracy and computational speed is significantly superior to the state-of-the-art classification methods.

V. Conclusion

This article proposed a novel SPMSFE method for the classification of HSIs. Three main steps were involved in the proposed SPMSFE method, i.e., identification of superpixel-based neighbors, multiple statistical FE, and multitask kernel sparse representation model for classification. In the first step, contextual spatial information was exploited accurately. The multiple statistical features extracted in the second step adequately explored the spatial geometry information, tight correlations between different spectral bands, and spatial–spectral variations from different perspectives. The MTKSRC model constructed in the third step offered a natural method to fuse the multiple statistical features for classification. The experimental results of three widely used HSI data sets indicated that the proposed SPMSFE method is superior to the state-of-the-art classification methods by a large margin, particularly when the training samples are very limited. Nevertheless, multiple statistical features extracted in the proposed SPMSFE play the same role for classification. Therefore, we would like to analyze the detailed effect of each statistical feature for HSI classification and then design appropriate multiple kernel learning methods to handle the heterogeneous fusion of multiple statistical features to further improve the classification performance in future works.

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